

Comparing K-Means and Other Algorithms for Data Clustering in Assembly.

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january 30, 2024

1 Purpose:

Some weeks ago I found the article: Comparing K-Means and Others Algorithms for Data Clustering by Nicolás Descartes. With C# source code.

Look very interesting to translate that to Assembly. Still is a work in progress but look well.

There is some kind of abuse of Collections, because is very easy to write that in C#. I removed most obvious exaggerations (perhaps author try to make more clear the algorithm, not sure).

- K-Means strategy need initial randomness, then probably you have to run several times before to find the better solution.
- Hierarchical strategy need to define termination, and in these cases are number of clusters.
- Density-based spatial clustering of applications with noise really benefit from Collections, and also need a couple of Sorted Vectors.

Anyway is a good challenge, and still I have to collect some leaks

2 Running the Program:

- Step 1: Select from Menu Data one data set. If you want to use default set or already loaded set, then skip this step.

- Step 2: Select from Menu File > Load Data. If you want to use already loaded set, then skip this step.
- Step 3: Select from Menu Strategy the algorithm you want. If you want to use default strategy or already select algorithm, then skip this step.
- Step 4 : Only if you selected K-Means or Hierarchical strategies. Select from Menu Cluster the number of clusters. In this examples the author use 3-clusters for data1 and data2, and 2-clusters for data 3.
- Step 5 : Select from Menu File > Run
- Step 6 : Go to Step1 to test more things.

3 Additional Reading

[Comparing K-Means and Others Algorithms for Data Clustering](#) by Nicolás Descartes